Week 9: matching the Medis centerlines to the outcomes

Progress

- **Starting point.** The trunk split and the progression analysis ended up with only about 200 patients with a Medis centerline tree, which made it look as if most of the 600 trees could not be linked to a study ID.
- **Matching only.** I wrote a short script that does nothing but the matching: the number in each tree's file name is the study ID, looked up in the baseline (BL) and follow-up (FU) spreadsheets.
- **All 600 trees matched.** Every tree has a patient in both the BL and FU sheets, no ID has more than one tree, and every scan was visually read at BL.
- **The loss is elsewhere.** The drop to about 200 happened in the trunk split, not in the ID matching. Most Medis files store the segment labels as bare codes, which the split did not translate, and their trunks were not found. With the codes translated, the split now uses all trees.
- **The trees come from diseased patients.** All but one of the 600 had plaque at BL, and most of them progressed (Table **??**).
- **What the trees can be used for.** The Medis trees are a reference from patients who already have plaque. They can validate the geometry from the in-house tracer and support a study of progression, but they cannot answer whether geometry predicts new plaque in the plaque-free cohort.

How the trunks were split

Each Medis tree was read as a graph of centerline points. For the LAD, LCx and RCA, the trunk is the path from the ostium (the LM bifurcation for the LAD and LCx) to the most distal point with that artery's label. The proximal, mid and distal boundaries were placed where the reader's Medis segment labels change along this path, and an unlabelled stretch between two labels was split at its middle. Medis gives the LCx after its proximal segment a single label, and this part was split into mid and distal at OM2 or the first side branch of at least 15 mm, otherwise halfway. Trunks without segment labels fell back to a landmark rule (first diagonal, OM1 and OM2, acute margin and crux). Pieces of a tree less than 3 mm apart were joined, and each artery was split on its own, so a failure in one artery left the other two in.

Table 1: Outcome of the 600 patients with a Medis tree, by plaque status at the first scan (rows) and what happened by the second scan (columns); row percentages in brackets. Plaque-free at BL means SIS = 0 and SSS = 0. Worse at FU means new plaque for a plaque-free patient, and for a patient with plaque that at least one segment has a higher stenosis grade than at BL. Counts from 1 to 4 are shown as < 5.

	At FU, 10 years later		
At BL	Worse	Not worse	Total
Plaque-free	< 5 new plaque	0 never plaque	< 5
Plaque	541 progressed (90 %)	58 stayed or regressed (10 %)	599
Total			600